1 The Group 17 elements are oxidising agents.

(a) **(i)** Explain how the Group 17 elements act as oxidising agents.

.....

..... [1]

(b) Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX.

$$3X_2 + 2NH_3 \rightarrow N_2 + 6HX \qquad X = Cl \text{ or } Br$$

The relative bond strengths of X–X and H–X determine the difference in enthalpy change of the two reactions.

(i) Describe and explain the difference in the X–X bond strengths of Cl_2 and Br_2.

.....

.....

.....

..... [2]

(v) I_2 reacts with NH_3 to form NI_3.

Predict the shape of a molecule of NI_3. Explain your answer.

shape .....

explanation .....

.....

[2]

3 Propan-2-ol, $(CH_3)_2CHOH$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

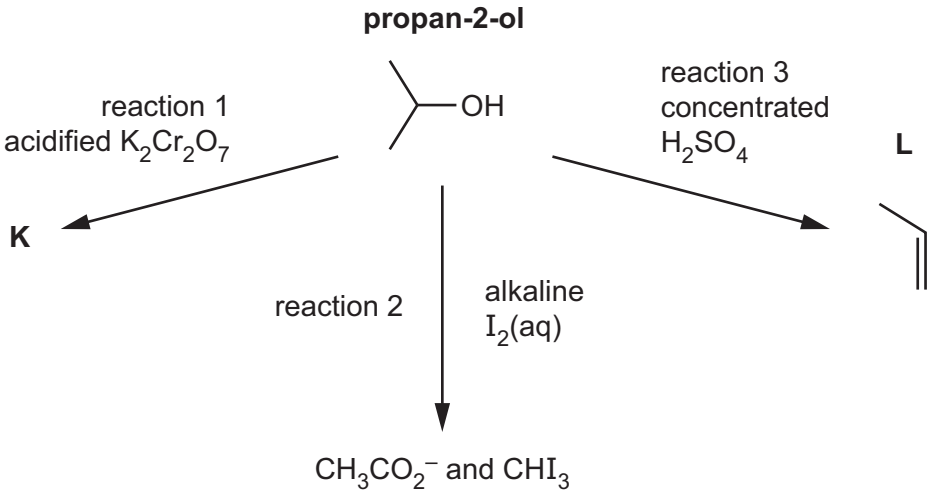


Fig. 3.1

(a) (i) Draw the structure of organic compound **K**.

[1]

(iv) Complete Fig. 3.2 to show the pi (π) bond in **L** that is formed from orbital overlap.

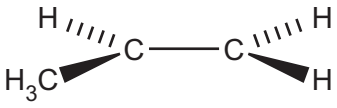


Fig. 3.2

[1]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(c) Acidified dichromate(VI) ions will convert methanal, CH_2O, to carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

(i) Identify the species that is reduced in half-equation 2. Explain your answer.

..... [1]

(iv) Methanal is a liquid at –30 °C but carbon dioxide is a gas at this temperature. Explain why.

.....

.....

.....

..... [2]

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) (i) Explain why the elements Na to A*l* are good electrical conductors.

.....

..... [1]

3 The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(a) Define homologous series.

.....

..... [2]

(d) Hexane, C_6H_{14}, has four structural isomers.

Fig. 3.1 shows hexane and two of its structural isomers.

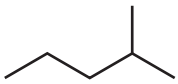
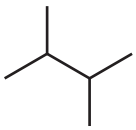
hexane	A	B	C	D
				

Fig. 3.1

(i) Complete Fig. 3.1 by drawing structures for **C** and **D**, the other two structural isomers of hexane. [2]

(ii) **A**, **B** and hexane have different boiling points.

Arrange **A**, **B** and hexane in order of increasing boiling point.

Explain your answer.

lowest ..... < ..... < ..... highest

.....

.....

.....
[3]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(c) Acidified dichromate(VI) ions will convert methanal, CH_2O, to carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

(i) Identify the species that is reduced in half-equation 2. Explain your answer.

..... [1]

(iv) Methanal is a liquid at –30 °C but carbon dioxide is a gas at this temperature. Explain why.

.....

.....

.....

..... [2]

2 **(a)** Fig. 2.1 shows the covalent bonds and lone pairs of electrons in a molecule of SCl_2.

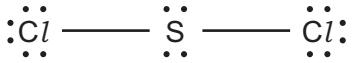


Fig. 2.1

(i) Use VSEPR theory to predict the shape of a molecule of SCl_2.

..... [1]

(ii) Predict the bond angle in a molecule of SCl_2.

..... ° [1]

3 $H_2(g)$ and $I_2(g)$ react to form HI(g) in a reversible reaction, as shown in equation 1.

equation 1 $\quad H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

(a) In experiment 1, $H_2(g)$ and $I_2(g)$ are combined.

Water is then added to the equilibrium mixture to produce 1.00 dm^3 of solution **A**.

The amount of $I_2(aq)$ present in 1.00 dm^3 of solution **A** is found by titration with $Na_2S_2O_3(aq)$.

Exactly 32.90 cm^3 of 0.200 mol dm^{-3} $Na_2S_2O_3$ reacts with all of the I_2 in a 25.0 cm^3 sample of solution **A**.

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

(i) Calculate the amount, in mol, of $S_2O_3^{2-}$ that reacts in the titration.

amount of $S_2O_3^{2-}$ = ... mol [1]

(d) The value of K_c for the dissociation of HCl at 298 K is 5.50×10^{-34}.

$$2HCl(g) \rightleftharpoons H_2(g) + Cl_2(g) \qquad K_c = 5.50 \times 10^{-34}$$

(i) Define covalent bond.

.....

..... [1]

(ii) Describe and explain the relative thermal stabilities of the hydrogen halides HCl and HI.

.....

.....

..... [2]

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

(a) A Period 3 oxide produces a solution with a pH greater than 10 when it is added to water. State the formula of the oxide.

..... [1]

(c) Table 2.1 shows the melting points of some oxides.

Table 2.1

oxide	melting point / °C
SO_2	–73
H_2O	0
SO_3	17
SiO_2	1610
Na_2O	1132
MgO	2852
Al_2O_3	2072

(i) Identify the oxide from Table 2.1 that contains the element with the highest oxidation number.

..... [1]

(ii) A student suggests the following hypothesis.

The melting point of an ionically bonded oxide is only determined by the charge on the cation.

Use Table 2.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

.....

.....

..... [2]

4 The reaction between $Na_2S_2O_3(aq)$ and $HCl(aq)$ is monitored at constant temperature.

$$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S + H_2O$$

Fig. 4.1 shows how the concentration of $HCl(aq)$ varies with time.

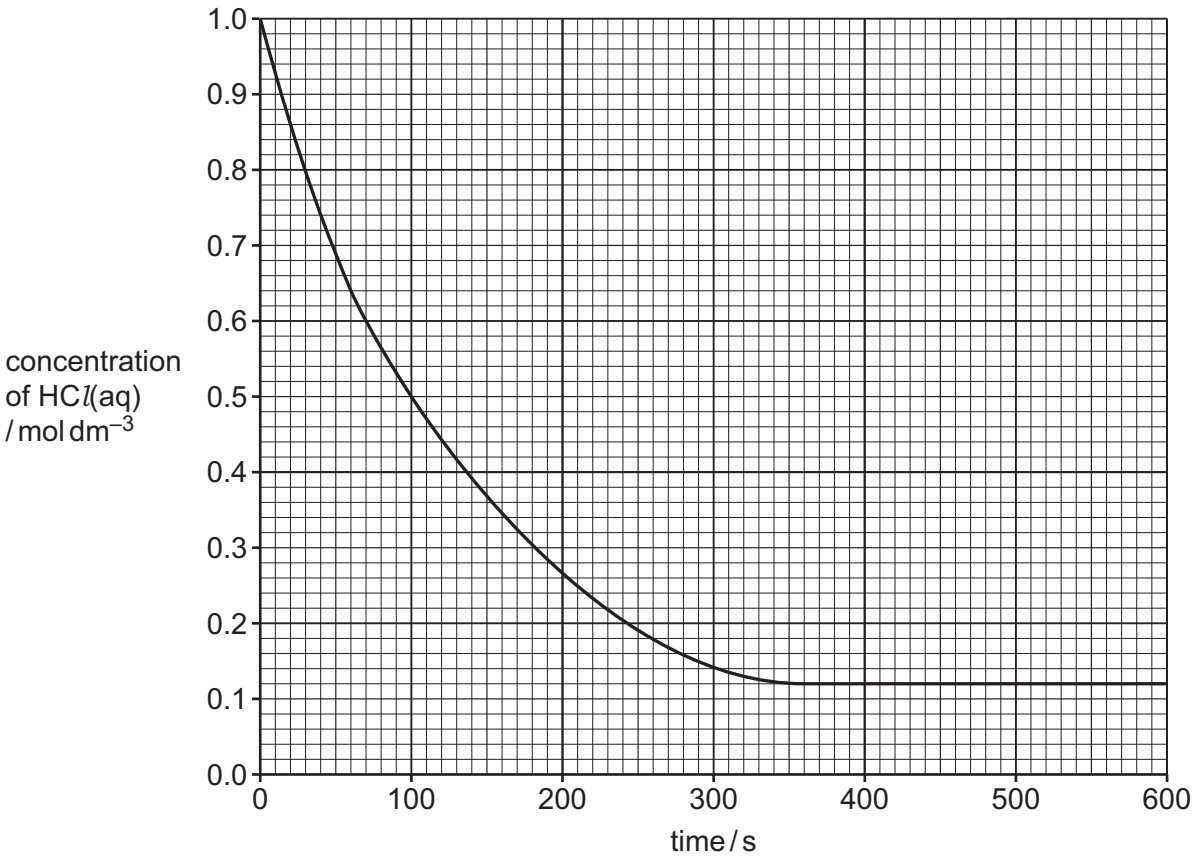


Fig. 4.1

(a) (i) Use Fig. 4.1 to find the average rate of change of concentration of $HCl(aq)$ in this reaction between 0–100 seconds and between 400–500 seconds. Include units in your answers.

0–100 seconds ..... units .....

400–500 seconds ..... units .....

[2]

(b) Fig. 4.2 shows a possible arrangement of outer-shell electrons in one SO_2 molecule.

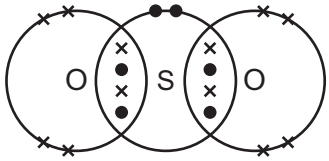


Fig. 4.2

(i) Use Fig. 4.2 to predict the shape and bond angle of a molecule of SO_2.

shape .....

bond angle .....°

[2]

(ii) Use Table 4.1 to predict the strength of the dipole moment of SO_2, if any, compared to that of H_2O. Explain your answer.

Table 4.1

	H	O	S
electronegativity	2.1	3.5	2.6

.....

.....

..... [2]

1 **(a)** Diamond and graphite are both crystalline solids made from carbon atoms. Graphite conducts electricity. Diamond does **not** conduct electricity.

(i) Name the type of lattice structure present in the crystalline solids diamond and graphite.

..... [1]

(d) The melting points of different oxides are shown in Table 1.1.

Table 1.1

oxide	melting point / °C	force of attraction broken during melting
SO_2	–73	
H_2O	0	
SiO_2	1610	
Na_2O	1132	
MgO	2852	

(i) Complete Table 1.1 by identifying the strongest force of attraction in each oxide that is broken during melting. Use the abbreviations below.

i.d. = instantaneous dipole–induced dipole

p.d. = permanent dipole–permanent dipole

H = hydrogen bond

C = covalent bond

I = ionic bond

[3]

(ii) A student suggests the following hypothesis.

The stronger the covalent bond between atoms in non-metal oxides, the higher the melting point.

Use Table 1.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

.....

.....

..... [2]

4 **(a)** Propanoic acid, CH_3CH_2COOH, reacts with reducing agent **Q** to produce propan-1-ol, $CH_3CH_2CH_2OH$.

(i) Suggest the formula of reducing agent **Q**.

..... [1]

(c) Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.

(i) Identify the type of hybridisation shown in the saturated carbon atoms of all alcohols.

..... [1]

1 **(a)** Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

(i) Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium .....

bonding in sodium oxide .....
[1]

(ii) Explain how solid sodium conducts electricity.

.....

..... [1]

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(a) Chlorine forms three different compounds with phosphorus.

The most common compounds are PCl_3 and PCl_5.

(i) Complete Table 1.1.

Table 1.1

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2.

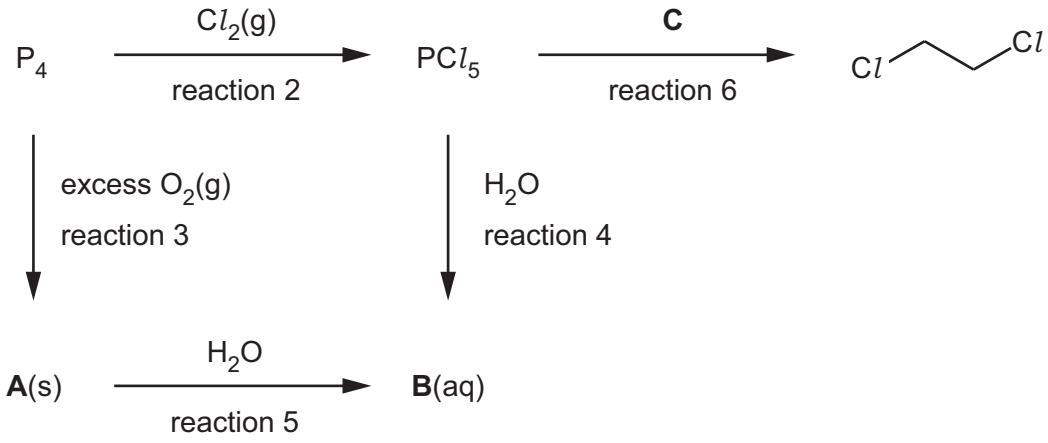


Fig. 1.1

(i) Suggest what is observed in reaction 2.

..... [1]

(ii) Predict the shape of a molecule of PCl_5.

..... [1]

2 **(a)** The Group 2 elements show metallic bonding.

Define metallic bonding.

.....

..... [2]

3 The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

(a) **(i)** Complete Table 3.1. [1]

(d) Iodine monobromide, IBr, is a dark red solid that melts near room temperature.

IBr reacts with propene. The mechanism for this reaction is the same as the mechanism for that of HBr with propene.

(i) Identify **all** the intermolecular forces that exist between molecules of IBr.

.....

..... [1]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(iii) Explain why Sn has good electrical conductivity.

.....

..... [1]

(d) Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

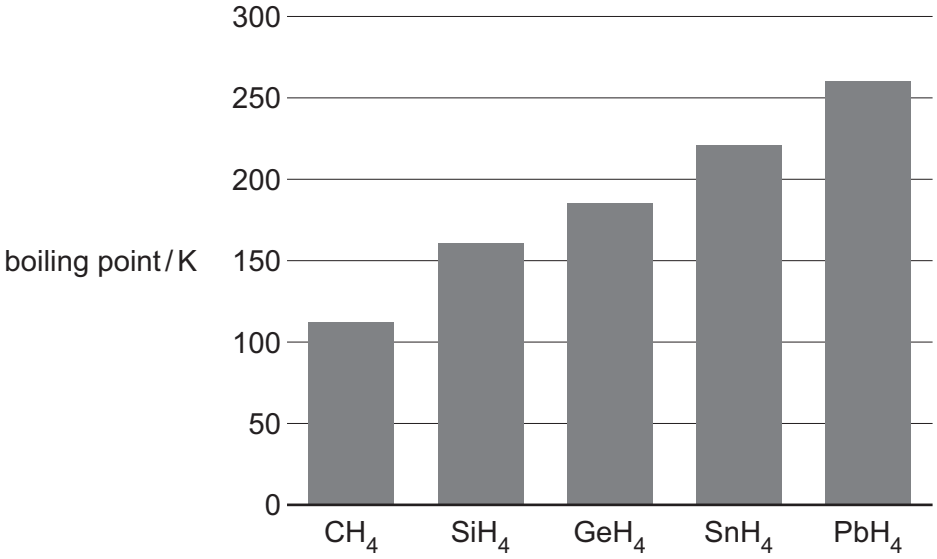


Fig. 3.2

(i) Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

.....

.....

.....

..... [2]

(ii) Deduce the shape of a molecule of SiH_4.

..... [1]

2 Oxygen is a Group 16 element.

(a) **(i)** Write equations for the following reactions.

- sodium and oxygen

.....

- sulfur and oxygen

.....

[2]

(ii) Draw a dot-and-cross diagram to show the species present in Al_2O_3.

Draw outer electrons only.

[1]

(c) Fig. 2.1 shows the boiling points of H_2O and other Group 16 hydrides.

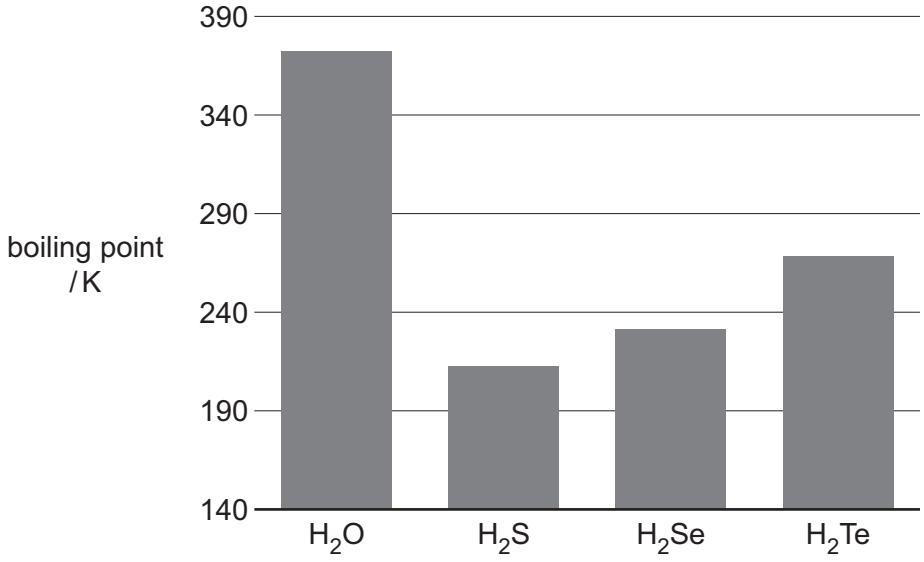


Fig. 2.1

(i) Explain the trend in the boiling points of the Group 16 hydrides H_2S to H_2Te.

.....

.....

.....

..... [2]

(ii) Explain why the boiling point of H_2O is much higher than that of H_2S.

.....

.....

..... [1]

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.

(i) Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural .....

man-made .....
[2]

(b) A common test for nitrates is the reaction with NaOH and A*l*. Equation 1 shows the reaction.

equation 1 $3NO_3^- + 8Al + 5OH^- + 18H_2O \rightarrow 3NH_3 + 8[Al(OH)_4]^-$

(i) Deduce the oxidation state of nitrogen in NO_3^-.

..... [1]

(iv) Suggest the shape of the $[Al(OH)_4]^-$ ion.

..... [1]

(f) At very high temperatures, phosphorus can form P_2 molecules.

P_2 contains a triple bond, P≡P.

(i) Describe the formation of the P≡P bond in terms of orbital overlap.

.....

.....

.....

..... [2]

(ii) The bond energy of P≡P is 485 kJ mol^{-1}. The bond energy of N≡N is 944 kJ mol^{-1}. Compare the reactivity of P_2 and N_2. Explain your answer.

.....

.....

..... [1]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(iii) Explain why Sn has good electrical conductivity.

.....

..... [1]

(d) Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

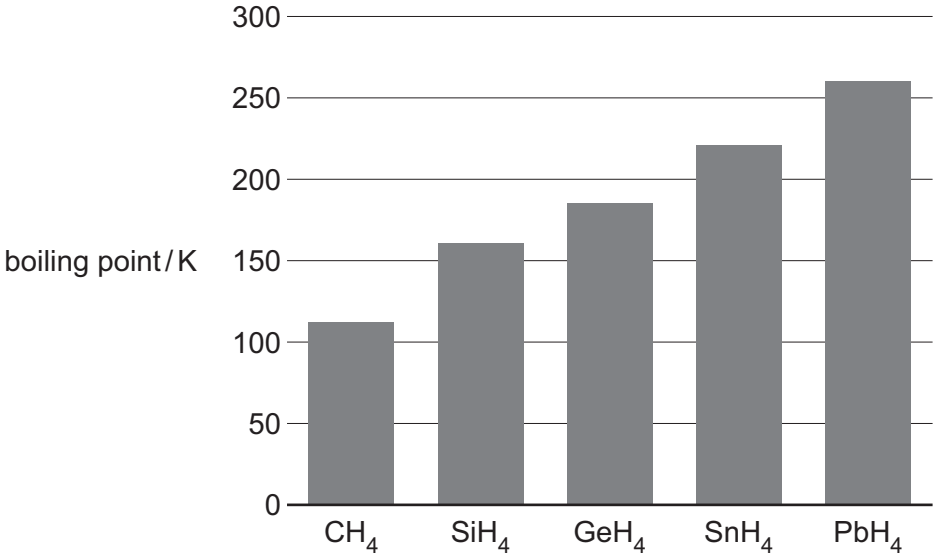


Fig. 3.2

(i) Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

.....

.....

.....

..... [2]

(ii) Deduce the shape of a molecule of SiH_4.

..... [1]

1 **(a)** **(i)** Explain the lack of reactivity of nitrogen gas, $N_2(g)$.

.....

.....

..... [2]

(ii) Covalent bonds can be σ bonds or π bonds.

Complete Table 1.1 to show the number of σ and π bonds in a molecule of N_2 and to describe how the orbitals overlap to form σ and π bonds.

Table 1.1

	σ bond	π bond
number of bonds in N_2		
how the orbitals overlap		

[4]

4 **(a)** $NH_3(g)$ reacts with $HCl(g)$ to produce $NH_4Cl(s)$, as shown.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl.

[2]

1 **(a)** Complete Table 1.1 using relevant information from the Periodic Table.

Table 1.1

	nucleon number	proton number	number of electrons
Mg^{2+}	24		
Al^{3+}	27		

[2]

(c) Draw a labelled diagram to show the structure and bonding in sodium.

[1]

3 **(a)** Describe what is meant by dynamic equilibrium.

.....

.....

..... [2]

(d) $SCN^-(aq)$ is colourless.

Complete the dot-and-cross diagram in Fig. 3.1 to show the arrangement of outer electrons in an SCN^- ion.

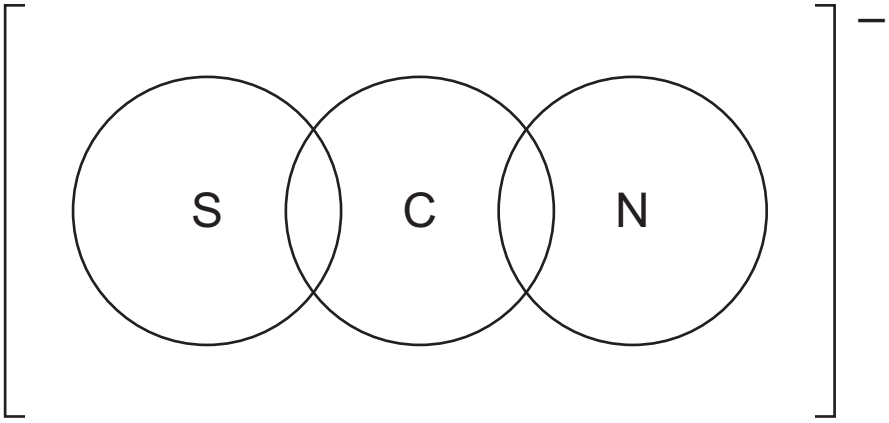


Fig. 3.1

[2]

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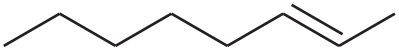
4 $CH_3(CH_2)_5CHBrCH_3$ exists as a pair of stereoisomers.

(a) Draw the three-dimensional structures of the **two** stereoisomers of $CH_3(CH_2)_5CHBrCH_3$. R can be used to represent $CH_3(CH_2)_5$.

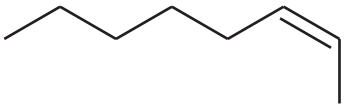
[2]

(d) $CH_3(CH_2)_5CHBrCH_3$ is heated with **D** to produce three different molecules, **E**, **F** and **G**.

E



F



G



Fig. 4.2

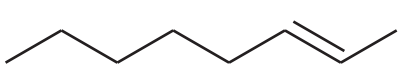
(i) Name the type of reaction.

..... [1]

(e) **(i)** Both σ and π bonds are present in a molecule of **E** as a result of different types of hybridisation in the carbon atoms.

Complete Table 4.2 to show the number of carbon atoms with each type of hybridisation in a molecule of **E**.

Table 4.2

	number of carbon atoms		
	sp hybridised	sp^2 hybridised	sp^3 hybridised
E 			

[2]

2 **(a)** Sulfur chloride, SCl_2, is a liquid at room temperature. When SCl_2 is added to water, misty fumes are seen and a solution is made that turns universal indicator red.

(i) Identify the type of reaction that occurs when SCl_2 is added to water.

..... [1]

(b) A molecule of SCl_2 contains two S–Cl covalent bonds.

(i) Complete the dot-and-cross diagram in Fig. 2.1 to show the arrangement of the outer electrons in a molecule of SCl_2.

Use × to show electrons from the chlorine atoms.
Use ● to show electrons from the sulfur atom.

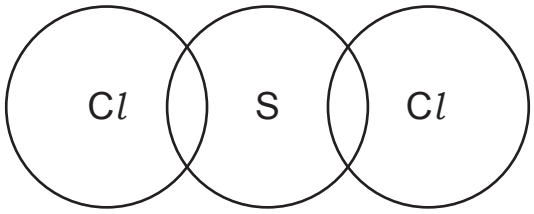


Fig. 2.1

[2]

(ii) Predict the shape of, and bond angle in, a molecule of SCl_2 by using VSEPR theory.

shape .....

bond angle .....

[2]

5 Hydrocarbon molecules contain covalent bonds.

(a) Define covalent bond.

.....

..... [1]

(b) A C=C bond in an alkene is made from a σ bond and a π bond.

(i) Identify the hybridisation of the carbon atoms in a C=C bond in an alkene.

..... [1]

(ii) Draw labelled diagrams to show, in terms of orbital overlap, how the σ and π bonds are made in a C=C bond.

σ bond

π bond

[2]

(c) In electrophilic reactions involving alkenes the π bond of C=C is broken.

(i) Suggest **one** difference between σ and π bonds that explains why the π bond is broken in electrophilic addition reactions involving alkenes.

.....

..... [1]

1 Bismuth is an element in Group 15 of the Periodic Table.

(a) Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

2 Chlorine, Cl_2, reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(a) Complete Table 2.1. [3]

(c) An excess of Cl_2 reacts with phosphorus to form PCl_5.

(i) PCl_5 is a simple molecule in the gas phase.

It also exists in a solid form as two ions, PCl_4^+ and PCl_6^-.

Complete Table 2.2 to identify the shapes of each of these species.

Table 2.2

species	PCl_5	PCl_4^+	PCl_6^-
shape		tetrahedral	

[2]

3 Nitrogen, N_2, is generally an unreactive molecule but it does react under certain conditions.

(a) Give **two** reasons to explain the lack of reactivity of nitrogen.

1 .....

2 .....

[2]

(d) The N_2F_2 molecule has a double covalent bond between its nitrogen atoms. This consists of a σ and a π bond.

(i) Complete Fig. 3.2 to show the dot-and-cross diagram for N_2F_2.

Show outer electrons only.

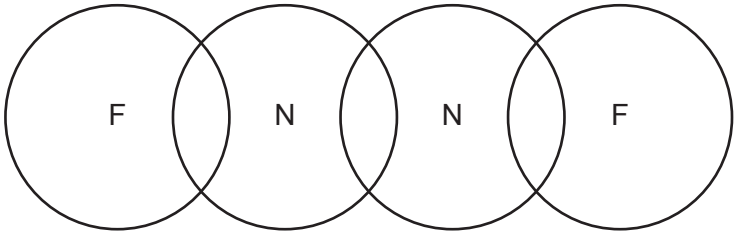


Fig. 3.2

[2]

(ii) Deduce the hybridisation of the N atoms in N_2F_2.

..... [1]

(iii) Draw a diagram of the π bond between the N atoms in N_2F_2 and describe how it forms.

.....

.....

[2]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) **(i)** Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

(d) $POCl_3$ shows similar chemical properties to PCl_5.

$POCl_3$ has a melting point of 1°C and a boiling point of 106 °C.

$POCl_3$ reacts vigorously with water, forming misty fumes and an acidic solution.

(i) Explain how the information in **(d)** suggests the structure and bonding of $POCl_3$ is simple covalent.

.....

.....

..... [2]

(iii) $POCl_3$ contains a double covalent bond between P and O.

Complete the dot-and-cross diagram, in Fig. 1.1, to show the bonding in $POCl_3$.

Show outer shell electrons only.

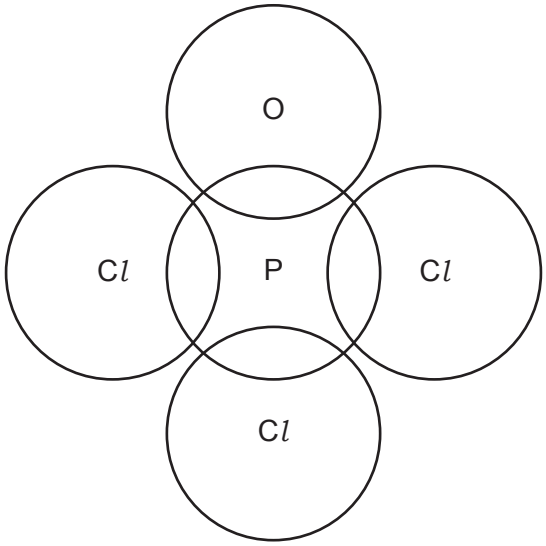


Fig. 1.1

[2]

1 The elements silicon, phosphorus and sulfur are in Period 3 of the Periodic Table.

(a) **(i)** Describe the variation in atomic radius from silicon to sulfur.

..... [1]

(b) Table 1.1 shows some properties of the elements Si to S.

The first ionisation energy of P is **not** shown.

Table 1.1

property	Si	P	S
total number of electrons in s subshells			
total number of electrons in p subshells			
first ionisation energy / kJ mol^{-1}	786		1000
formula of most common chloride	$SiCl_4$	PCl_5	SCl_2

(i) Complete Table 1.1 to show the total number of s and p electrons in an atom of Si, P and S.

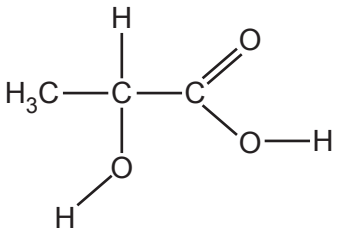
[2]

(v) Predict the shape of the SCl_2 molecule.

..... [1]

4 Lactic acid, $CH_3CH(OH)COOH$, and pyruvic acid, $CH_3COCOOH$, both contain two functional groups.

lactic acid



pyruvic acid

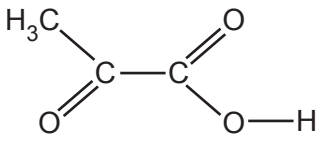


Fig. 4.1

(a) (i) Explain why lactic acid exists as optical isomers.

.....

.....

..... [1]

(iii) Lactic acid forms hydrogen bonds with water.

Complete Fig. 4.2 to show the formation of a hydrogen bond between one molecule of lactic acid and one molecule of water.

Label the hydrogen bond. Show any relevant dipoles and lone pairs of electrons.

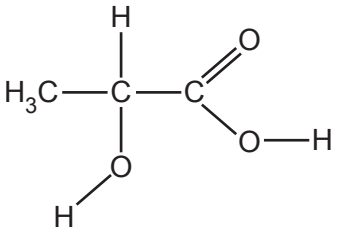


Fig. 4.2

[3]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) **(i)** Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

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$POCl_3$ has a melting point of 1°C and a boiling point of 106 °C.

$POCl_3$ reacts vigorously with water, forming misty fumes and an acidic solution.

(i) Explain how the information in **(d)** suggests the structure and bonding of $POCl_3$ is simple covalent.

.....

.....

..... [2]

(iii) $POCl_3$ contains a double covalent bond between P and O.

Complete the dot-and-cross diagram, in Fig. 1.1, to show the bonding in $POCl_3$.

Show outer shell electrons only.

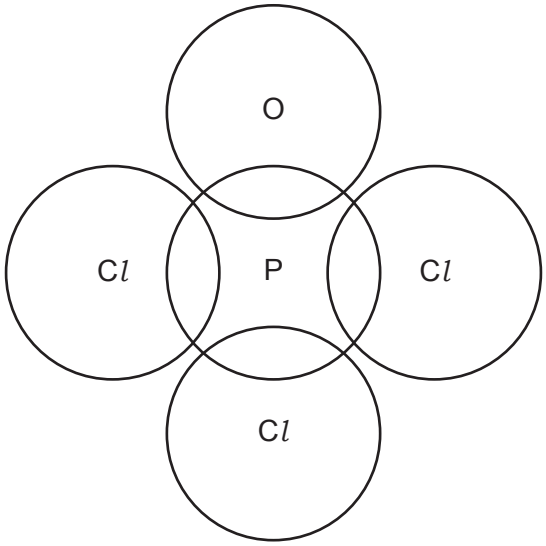


Fig. 1.1

[2]

1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

(a) **(i)** Identify the lattice structure of copper.

..... [1]

(ii) Draw a labelled diagram to show the bonding present in copper.

[1]

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.



Fig. 5.1

(a) Deduce the empirical formula of **Y**.

..... [1]

(c) Complete Table 5.1 to show the number of sp^2 and sp^3 hybridised carbon atoms in a molecule of **X**.

Table 5.1

type of hybridisation	sp^2	sp^3
number of carbon atoms in **X**		

[2]

1 The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point / K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

(a) (i) State the type of bonding present in magnesium and in sodium chloride.

bonding in magnesium .....

bonding in sodium chloride .....
[1]

(ii) Explain the difference in the melting points of magnesium and sodium chloride.

.....

..... [1]

(iii) Explain the difference in the melting points of phosphorus and sulfur in terms of structure and bonding.

.....

.....

..... [2]

(b) (i) Define electronegativity.

.....

..... [1]

(ii) Explain why electronegativity increases across a period.

.....

.....

..... [2]

(iii) Name the strongest intermolecular force that exists between NH_3(l) molecules.

..... [1]

(iv) Draw a diagram to show the formation of the strongest intermolecular force between **two** molecules of $NH_3(l)$.

Include any relevant lone pairs of electrons and dipoles.

[2]

(v) The melting points of ice and ammonia are shown in Table 1.2.

Table 1.2

solid	melting point / K
ice	273
ammonia	195

Suggest **two** reasons for the difference in the melting points of ice and ammonia.

.....

.....

..... [2]

3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from CH_3CH_2Cl.

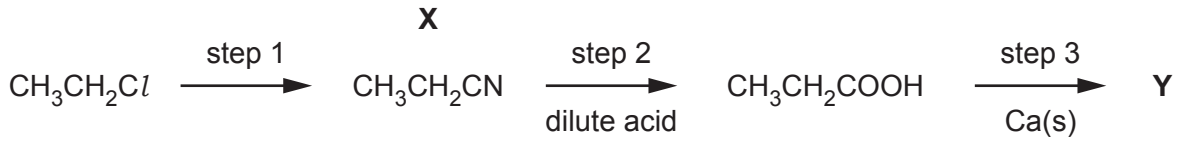


Fig. 3.1

(a) (i) State the reagent and conditions for step 1 in Fig. 3.1.

..... [1]

(ii) Give the systematic name of **X**.

..... [1]

(c) Complete Table 3.1 to show the number of sigma bonds (σ) and pi bonds (π) present in a molecule of **X**.

Table 3.1

type of bond	number of bonds in **X**
sigma (σ)	
pi (π)	

[2]

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4.

reaction 1 $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$

(a) Use the bond energy values in Table 4.1 to calculate the enthalpy change, ΔH_r, for reaction 1.

Table 4.1

bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
C–H	410
C–O	360
C=O	740
O–H	460

ΔH_r = ... kJ mol^{-1} [2]

(d) Describe the covalent bonds present between the carbon atoms in an ethene molecule by completing Table 4.2.

Table 4.2

	sigma (σ)	pi (π)
type of orbitals involved in bond		
how the orbitals overlap		

[2]

1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te. Its electronic configuration is [Kr] $4d^{10}\ 5s^2\ 5p^4$.

(a) Complete Table 1.1.

Table 1.1

	nucleon number	number of neutrons	number of electrons
^{130}Te			

[3]

(e) Te reacts with F_2 at 150 °C to form TeF_x. Molecules of TeF_x are octahedral with bond angles of 90°.

Explain why TeF_x is octahedral with bond angles of 90°.

.....

.....

..... [2]

4 **V** is a colourless liquid.

V

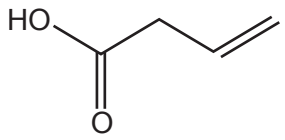


Fig. 4.1

(a) **V** reacts with an excess of $LiAlH_4$ to form **W**.

(i) Draw the structure of **W** in the box.

W

[1]

(iii) Complete Table 4.1 to show the number of sp^2 and sp^3 hybridised carbon atoms that are present in a molecule of **V**.

Table 4.1

type of hybridisation	sp^2	sp^3
number of carbon atoms in **V**		

[2]

1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy / kJ mol^{-1}	1310	519	1090	1310	1000
second ionisation energy / kJ mol^{-1}	—	7300	2350	3390	2260

(a) (i) Define electronegativity.

.....

..... [1]

(ii) O and S are in Group 16.

Explain the difference in the Pauling electronegativity values of O and S.

.....

.....

..... [2]

(b) (i) LiH is an ionic compound.

Draw a dot-and-cross diagram of LiH.

Include **all** electrons.

[2]

(ii) Suggest the shape of a molecule of H_2S.

..... [1]

(e) **(i)** Complete Table 1.2 by placing a tick (✓) to show which of the compounds have molecules with an overall dipole moment.

Table 1.2

compound	O=C=O	O=S=O	S=C=S	S=C=O
overall dipole moment				

[2]

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(a) **(i)** Draw a labelled diagram to show the bonding and structure of the Group 2 metals at room temperature.

[2]

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2.

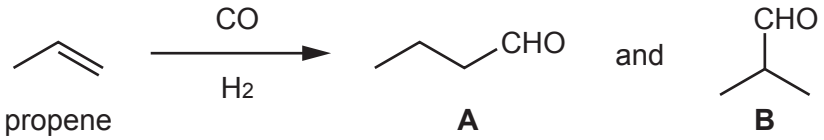


Fig. 3.1

(a) **(i)** Define addition reaction.

.....

..... [1]

(d) The reaction of ethene, C_2H_4, with a 1:1 mixture of CO and H_2 is shown in equation 1.

equation 1 $C_2H_4(g) + CO(g) + H_2(g) \rightleftharpoons CH_3CH_2CHO(g)$

At atmospheric pressure a cobalt-based catalyst is used in this reaction.

(i) State and explain the effect of using a catalyst on this reaction.

.....

.....

.....

..... [2]

(iv) The reaction mixture is cooled to collect CH_3CH_2CHO as a liquid.

Identify all types of van der Waals' forces that are present between molecules of CH_3CH_2CHO.

.....

..... [1]